Dijkstra's Algorithm for undirected graph

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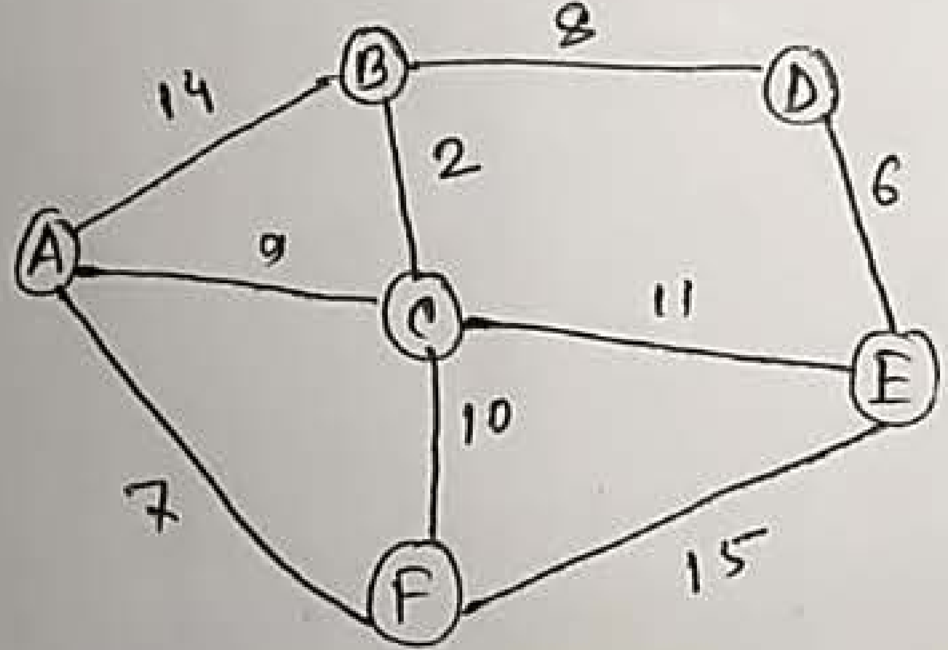
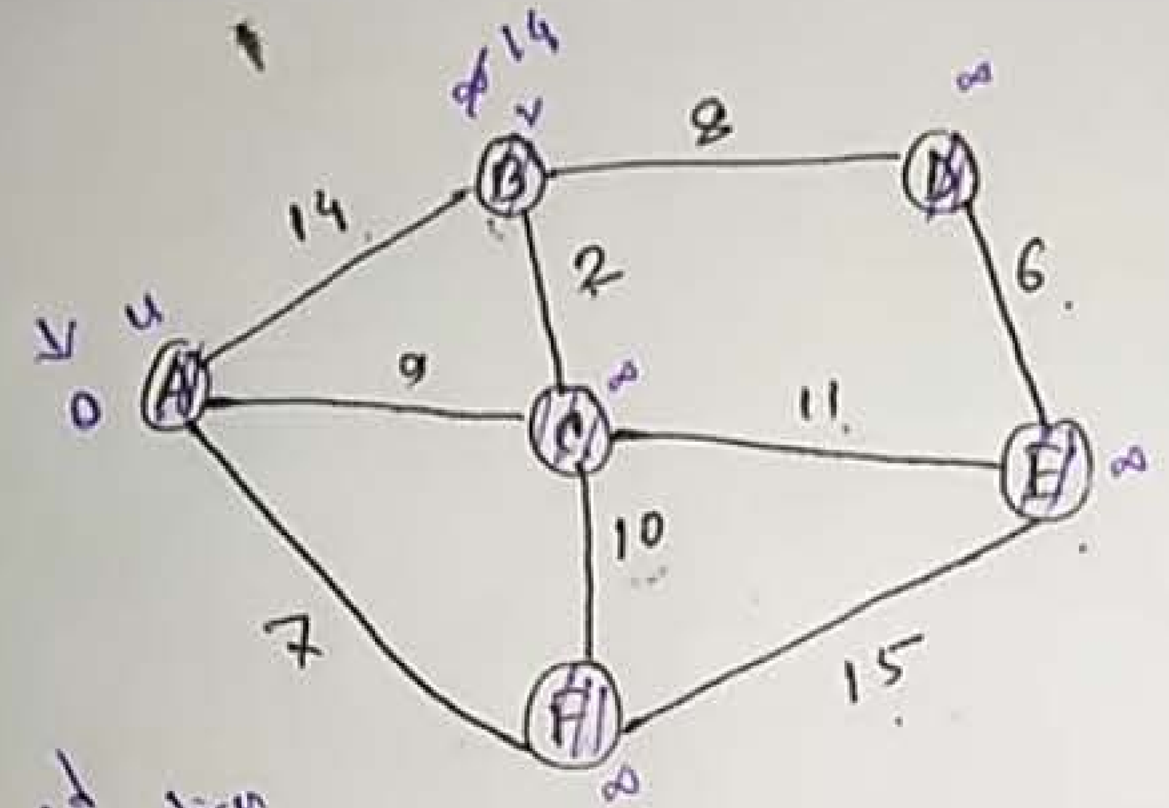
$\Rightarrow$ If $(d(u) + c(u,v) < d(v))$

$$d(v) = d(u) + c(u,v)$$



	A	B	C	[illegible]	E



Visited vertices

	3		2		1	
↓	A	B	C	D	E	F
A	$\boxed{0}$	α	α	α	α	α
F		14	9	α	α	$\boxed{7}$
C		14	$\boxed{9}$	α	22	
B		$\boxed{11}$		α	20	
D				$\boxed{19}$	20	
E					$\boxed{20}$	

(change)

go backward until you find change, then start from the "selected" / "boxed" in that row

A to C = 9
C to E = 11

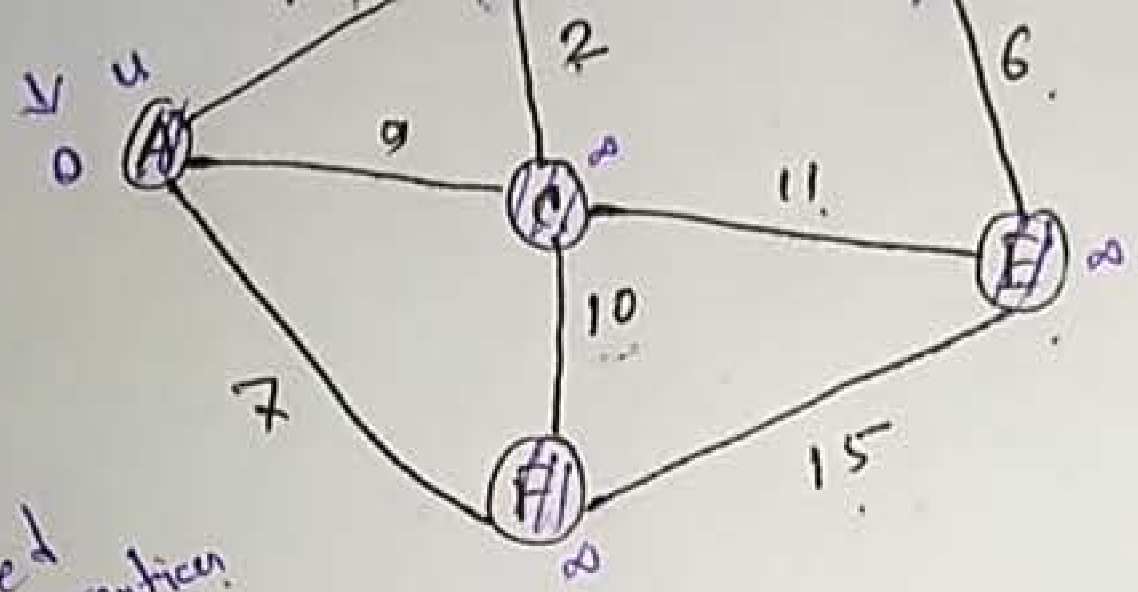
find:

A → E

⇒ E, C, A

A, C, E

$9 + 11 = 20$



visited vertices

	A	B	C	D	E	F
A	$\boxed{0}$	∞	∞	∞	∞	∞
F		14	9	∞	∞	$\boxed{7}$
C		14	$\boxed{9}$	∞	22	
B		$\boxed{11}$		∞	20	
D				$\boxed{19}$	20	
E					$\boxed{20}$	

A → D

D, B, C, A

A, C, B, D

9 + 2 + 8

Dijkstra's Algorithm for directed graph

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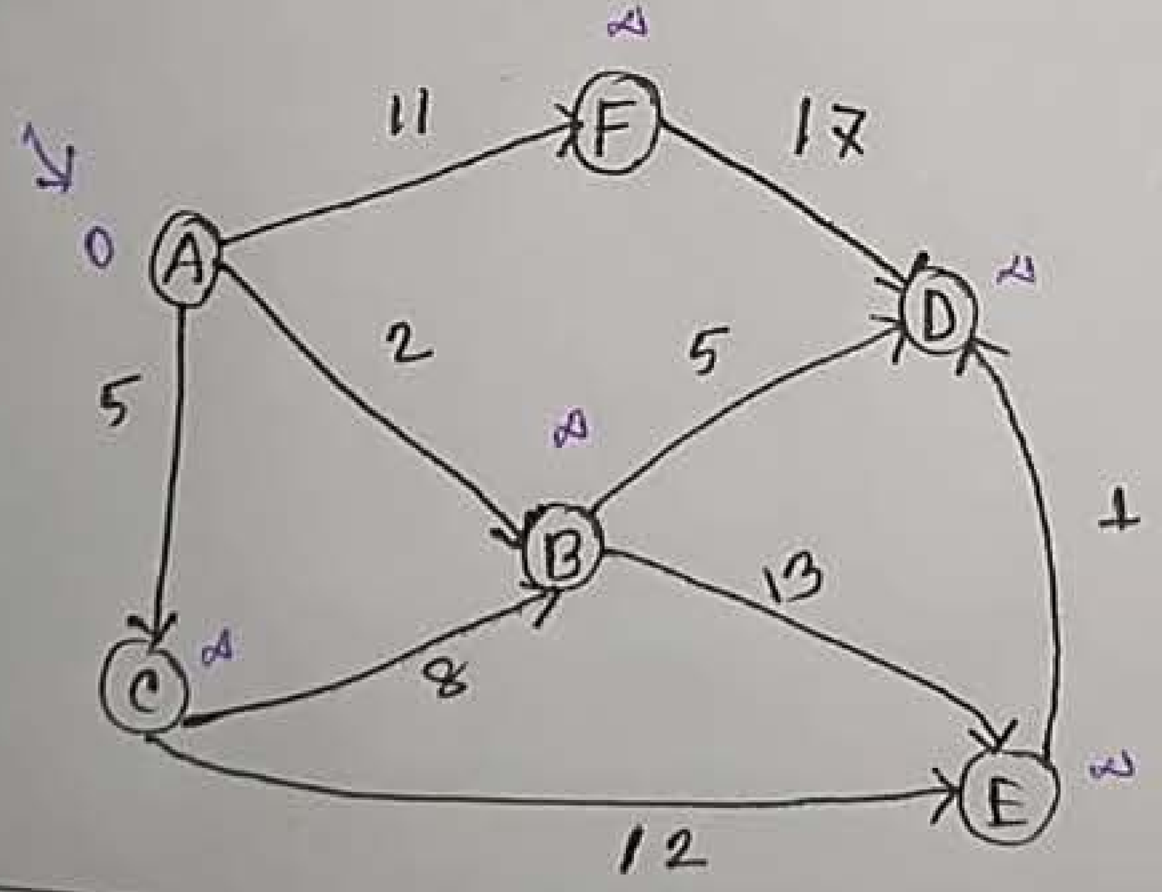
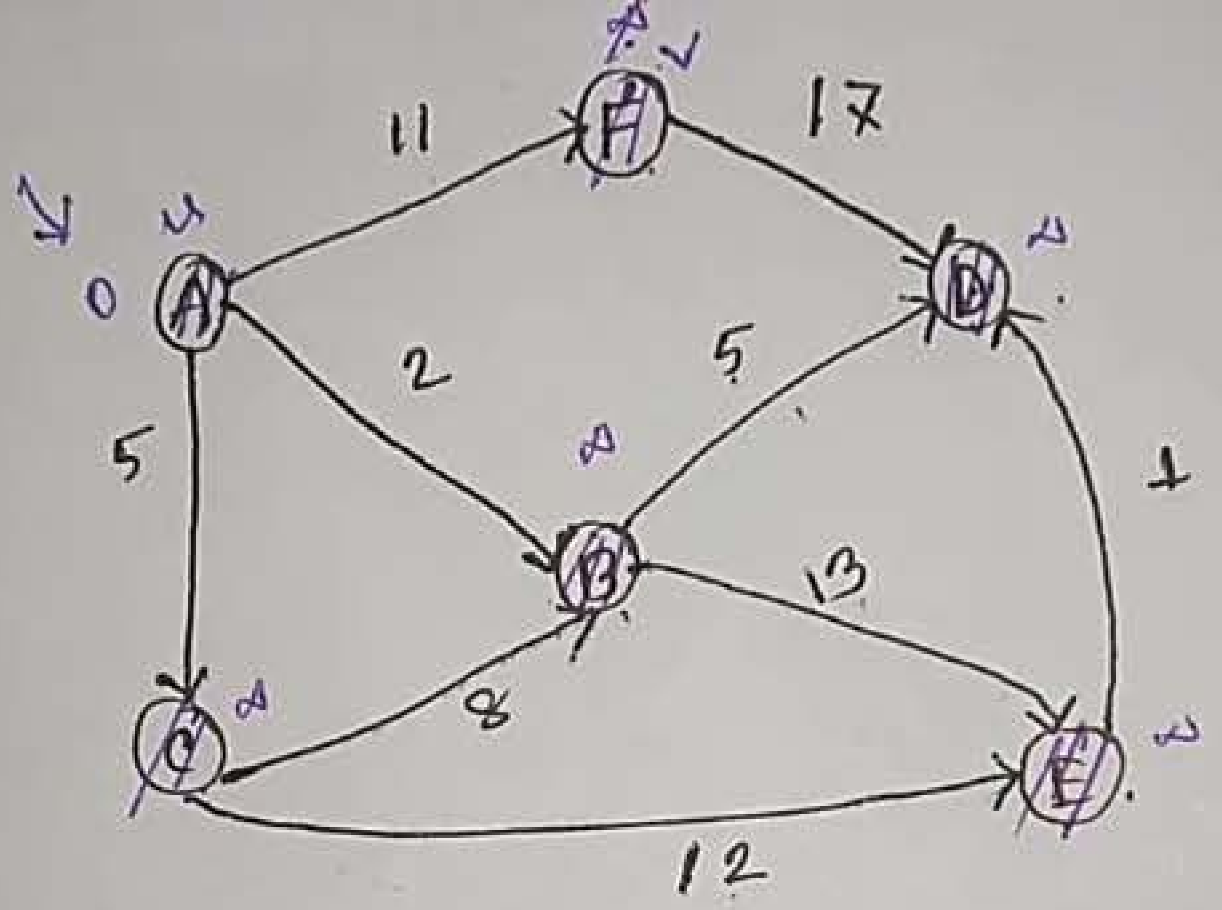
$\Rightarrow$ If $(d(u) + c(u,v) < d(v))$

$$d(v) = d(u) + c(u,v)$$



visited vertices

	A	B	C	D	E	F



visited vertices.

	A	B	C	D	E	F
A	$\boxed{0}$	α	α	α	α	α
B		$\boxed{2}$	5	α	α	11
C			$\boxed{5}$	7	15	11
D				$\boxed{7}$	15	11
F					15	$\boxed{11}$
E					$\boxed{15}$	

A → E

E, B, A

A, B, E

$2 + 13 = 15$